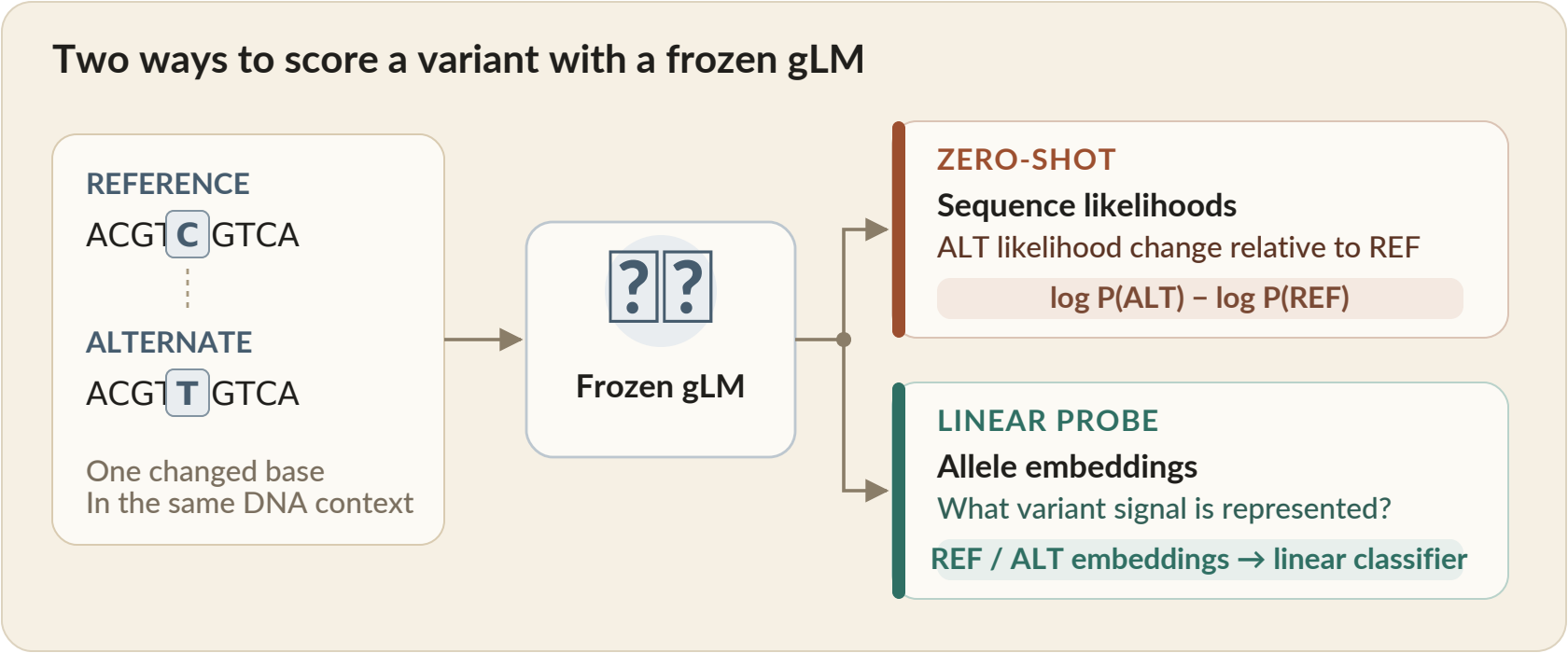
Two ways to score a variant with a frozen gLM
REFERENCE
ACGTCGTCA
ALTERNATE
ACGTTGTCA
One changed base
In the same DNA context
Frozen gLM
ZERO-SHOT
Sequence likelihoods
ALT likelihood change relative to REF
log P(ALT) – log P(REF)
LINEAR PROBE
Allele embeddings
What variant signal is represented?
REF / ALT embeddings → linear classifier